

**This assignment will not be graded and is only for practice.**

**Level 1:**

1) Consider a square matrix  $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$ . If **1 point**

$$P = -1M_{11} - 2M_{12} + 0M_{13}$$

$$Q = -2M_{21} + 0M_{22} + 1M_{23}$$

$$R = -1M_{31} - 0M_{32} + 1M_{33}$$

where  $M_{ij}$  is the minor with respect to the  $(i, j)$ -th entry, then choose the set of correct options.

☐  $\det(A) = -2$ .

☐  $P = Q$ .

☐  $P = Q = R$ .

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☐  $Q = R$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\det(A) = -2$ .

$P = Q = R$ .

2) Suppose for a real  $3 \times 3$  matrix  $A$ , there exists a real  $3 \times 3$  matrix  $P$  such that  $D = PAP^{-1}$  is a real  $3 \times 3$  diagonal matrix. Choose the correct set of options. **1 point**

☐  $\det(A)$  must be equal to  $\det(P)$ .

☐  $\det(A)$  must be equal to  $\det(D)$ .

☐ The matrix  $A$  must be equal to  $D$ .

☐ If  $D$  is the identity matrix of order 3, then  $A$  must be the identity matrix of order 3

No, the answer is incorrect.

Score: 0

## Accepted Answers:

$\det(A)$  must be equal to  $\det(D)$ .

If  $D$  is the identity matrix of order 3, then  $A$  must be the identity matrix of order 3

3) Choose the set of correct options for square matrices of order 3.

**1 point**

- ☐ If  $AB = 0$ , then one of the matrices between  $A$  and  $B$  must have determinant 0.
- ☐ If  $AB = 0$ , then both the matrices  $A$  and  $B$  must have determinant 0.
- ☐ If  $AB = 3I$ , where  $I$  denotes the identity matrix of order 3, then  $\det A = \frac{3}{\det(B)}$ .
- ☐ If  $AB = 3I$ , where  $I$  denotes the identity matrix of order 3, then  $\det A = \frac{9}{\det(B)}$ .
- ☐ If  $AB = 3I$ , where  $I$  denotes the identity matrix of order 3, then  $\det A = \frac{27}{\det(B)}$ .
- ☐ If  $AB = 3I$ , where  $I$  denotes the identity matrix of order 3, then both  $A$  and  $B$  must have non-zero determinant.

No, the answer is incorrect.

Score: 0

## Accepted Answers:

If  $AB = 0$ , then one of the matrices between  $A$  and  $B$  must have determinant 0.

If  $AB = 3I$ , where  $I$  denotes the identity matrix of order 3, then  $\det A = \frac{27}{\det(B)}$ .

If  $AB = 3I$ , where  $I$  denotes the identity matrix of order 3, then both  $A$  and  $B$  must have non-zero determinant.

4) Consider a matrix  $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ . If  $a + b + c$  is divisible by 6 then choose the set of correct options.

**1 point**

- ☐  $\det(A)$  is divisible by 5.
- ☐  $\det(A)$  is divisible by 6.
- ☐  $\det(A)$  is divisible by 2.

☐  $\det(A)$  is divisible by 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\det(A)$  is divisible by 6.

$\det(A)$  is divisible by 2.

$\det(A)$  is divisible by 3.

5) Let  $A = [a_{ij}]$  be a square matrix of order 3, where  $a_{ij} = i$ . Find  $\det(A)$ .

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

**1 point**

**Level 2:**

6) Let  $A = [\delta_{ij}]$  be a square matrix of order 3, where  $\delta_{ij} = i \times j$ . Find  $\det(A)$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

7) Consider the following square matrices:

1 point

$$A = \begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \\ 2019 & 2020 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$$

Choose the set of correct options.

- ☐  $\det(B) = \det(A)$ .
- ☐  $\det(A) = -\det(B)$ .
- ☐  $\det(C) = -2\det(B)$ .
- ☐  $\det(C) = -2\det(A)$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\det(A) = -\det(B).$$

$$\det(C) = -2\det(B).$$

$$\det(C) = -2\det(A).$$

(Based on the below information answer questions 8 and 9).

Let  $A$  be a real  $3 \times 3$  matrix as given below:

$$A = [C_1 \quad C_2 \quad C_3]$$

where  $C_i$  represents the  $i$ -th column of the matrix  $A$ . Now let us consider the following set of matrices.

$$A_1 = [C_1 \quad C_2 + 5C_3 \quad C_3]$$

$$A_2 = [C_1 + C_2 + C_3 \quad C_2 \quad C_3]$$

$$A_3 = [C_1 \quad C_2 + 5C_3 \quad 0]$$

$$A_4 = [C_1 + C_2 \quad C_2 + C_3 \quad C_3 + C_1]$$

8) Which of the matrices have the same determinant as that of  $A$ ?

**1 point**

☐  $A_1$

☐  $A_2$

☐  $A_3$

☐  $A_4$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$A_1$

$A_2$

9) Choose the set of correct options.

**1 point**

☐  $\det(A_3)$  cannot be determined from the given information.

☐  $\det(A_3) = 0$

☐  $\det(A_4) = 2 \det(A)$

☐  $\det(A_2) = 3 \det(A)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\det(A_3) = 0$

$\det(A_4) = 2 \det(A)$

10) If the diagonal entries of a  $3 \times 3$  lower triangular matrix  $A$  are 1, 2, and 3, then find the sum of roots of the equation  $\det(A - xI) = 0$ , where  $I$  is the identity matrix of order 3.



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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

***1 point***

**Your score is: 0/10**